Lattice-Based Accumulator and Application to Anonymous Credential Revocation

Victor Youdom Kemmoe Anna Lysyanskaya Ngoc Khanh Nguyen

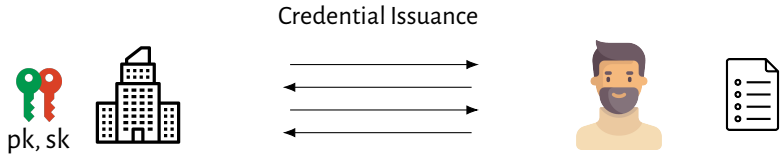


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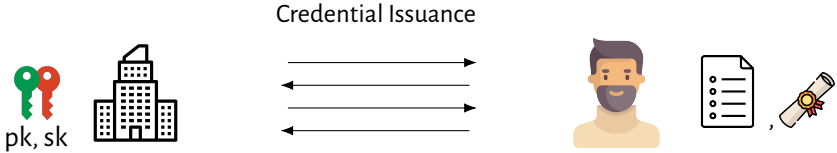
Motivation

Anonymous Credentials [CL02 a; BBC+24]



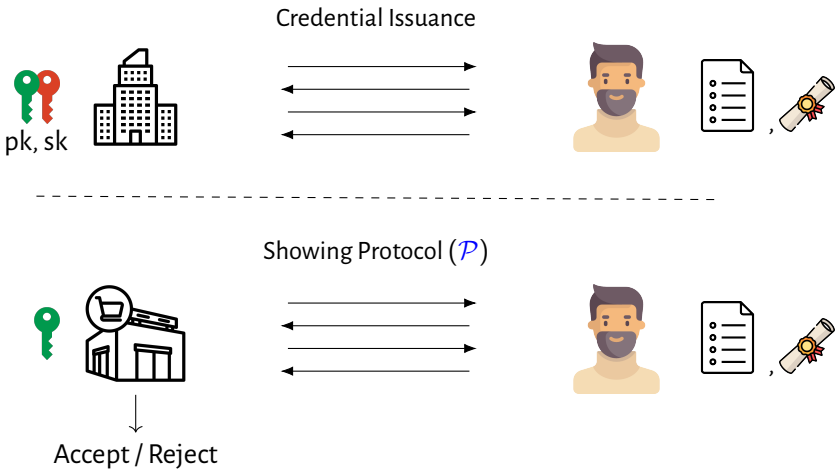
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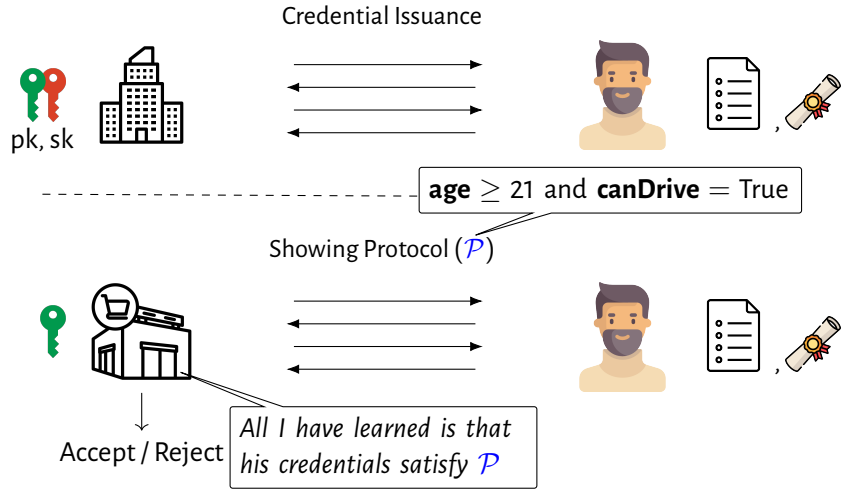
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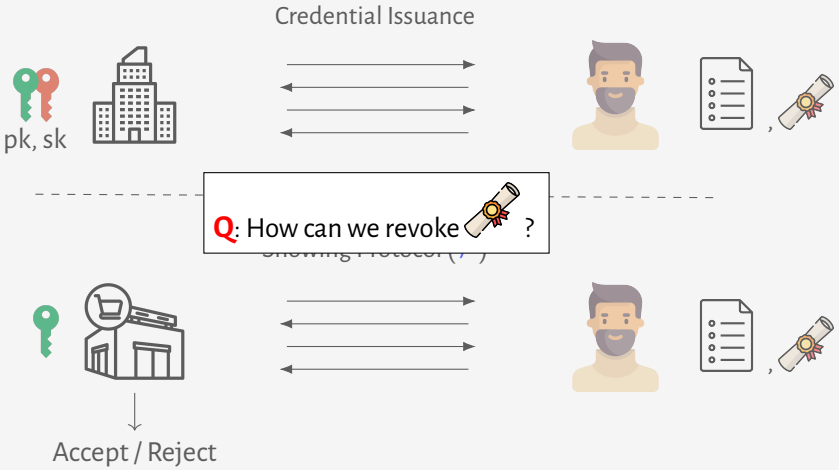
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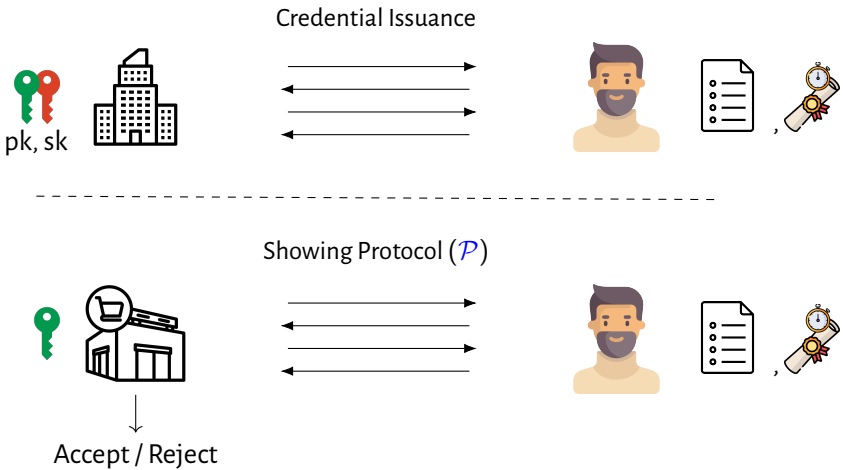
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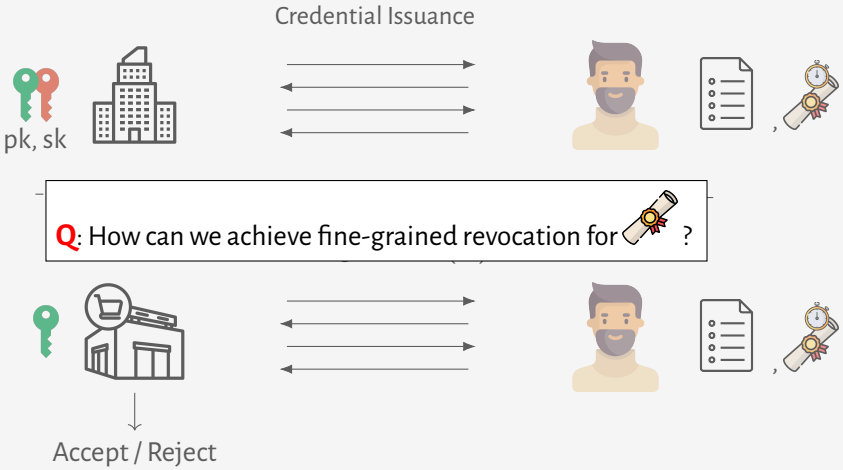
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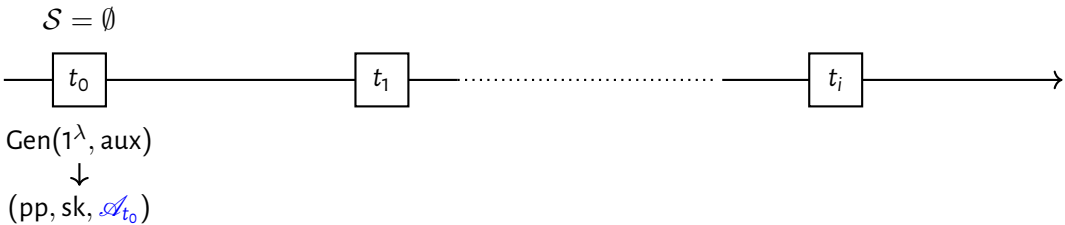
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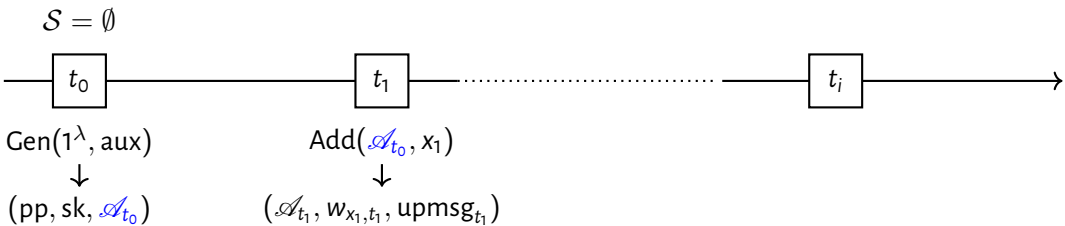
Positive Dynamic Accumulator

Syntax [BCD+17 ; DHS15 ; KL24]



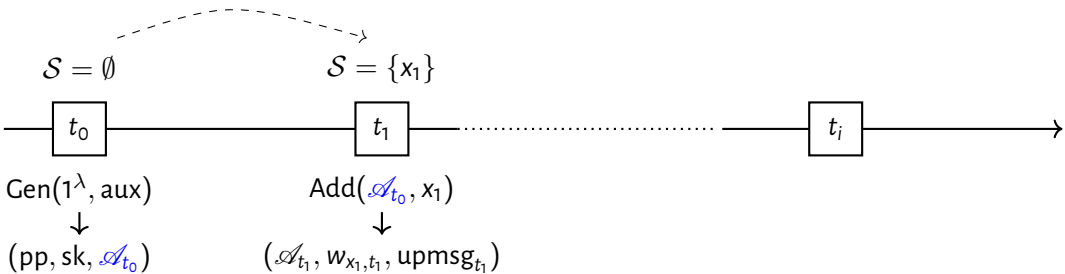
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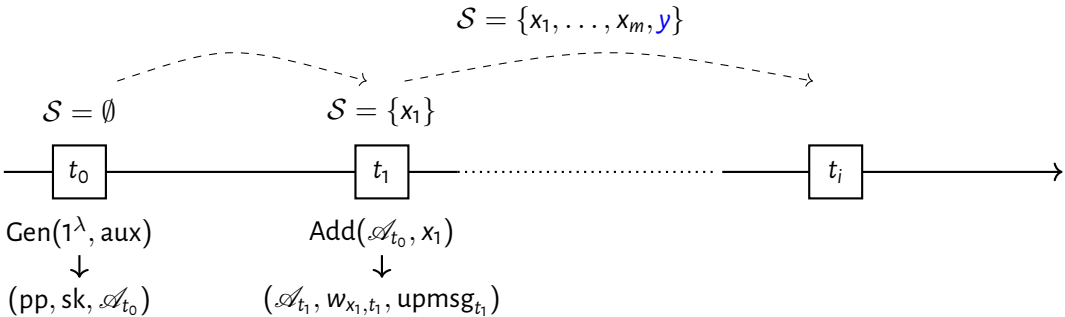
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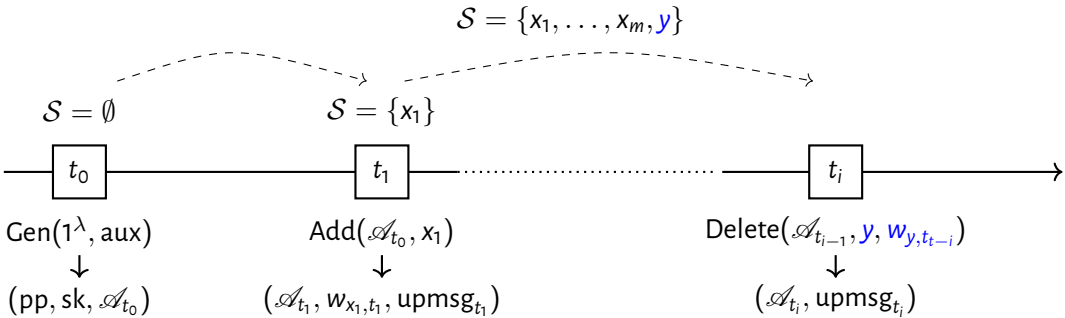
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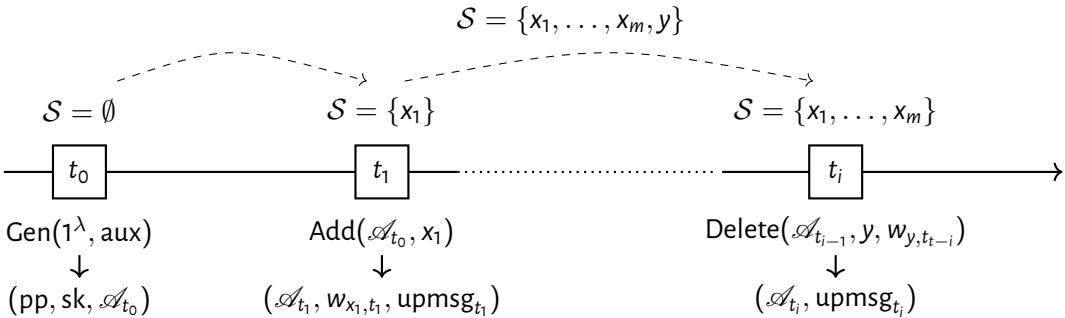
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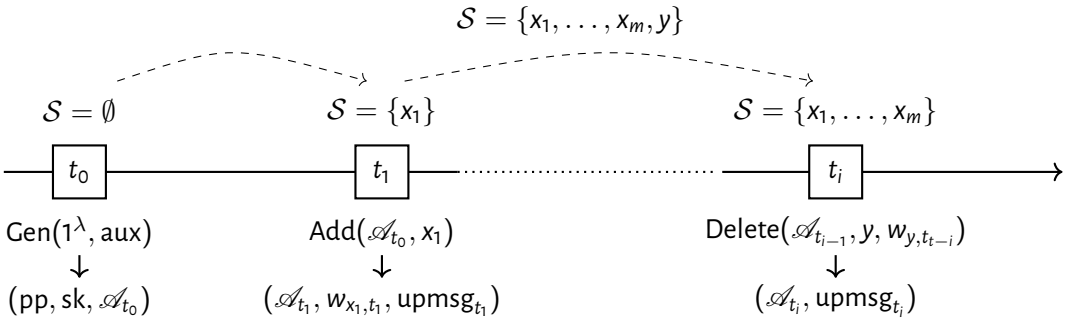
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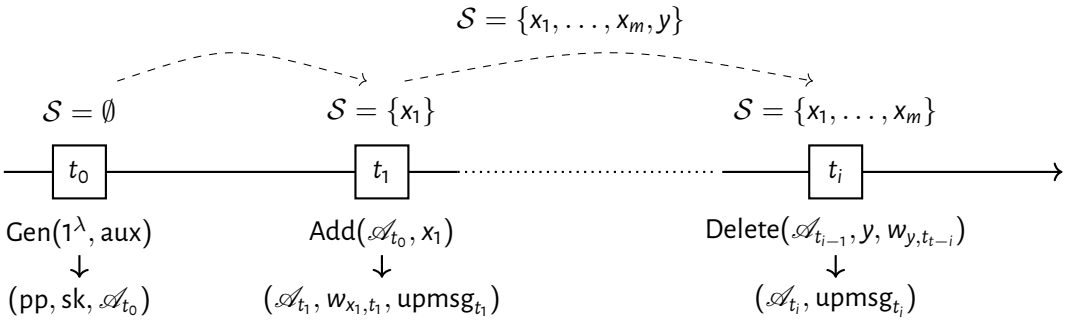
Syntax [BCD+17 ; DHS15 ; KL24]



- $\mathsf{MemWitUp}(x, w_{x,t}, \mathsf{upmsg}_{t+1}) \to w_{x,t+1}$
- $\mathsf{MemVerify}(\mathscr{A}_t, x, w_{x,t}) \to \mathsf{Accept}/\mathsf{Reject}$

Positive Dynamic Accumulator

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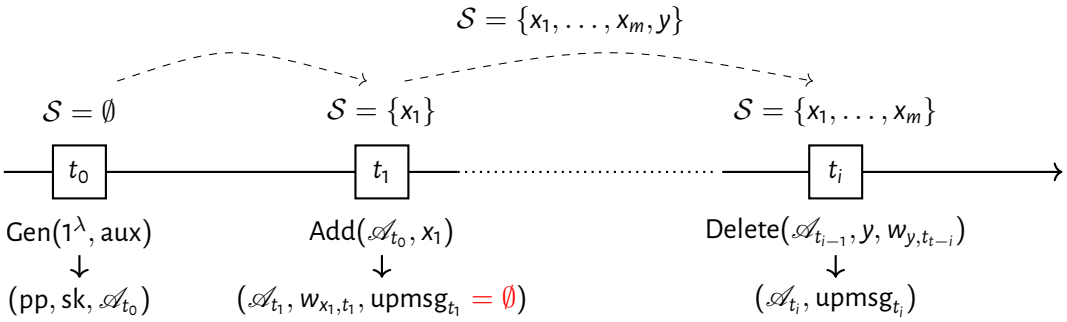


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- **Compactness**: $|\mathscr{A}| = \mathrm{poly}(\lambda), |w_{x,t}| = \mathrm{poly}(\lambda, |x|)$
- **Security**: *Hard* to produce a w_x for $x \notin \mathcal{S}$
- **Communication efficiency**: $|\mathsf{upmsg}| = O(\#\mathsf{Del})$

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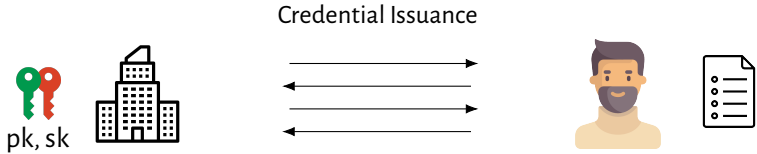
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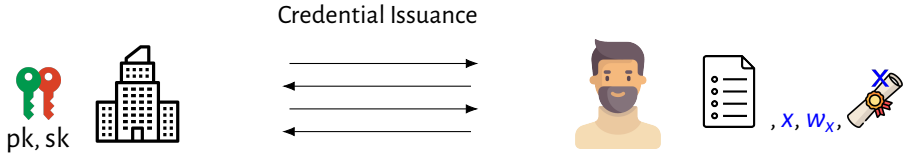
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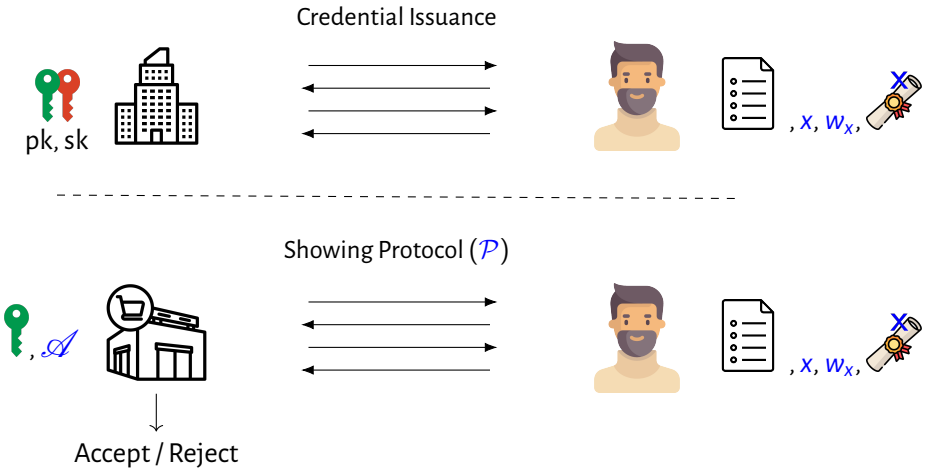
Positive Dynamic Accumulator in ACs revocation



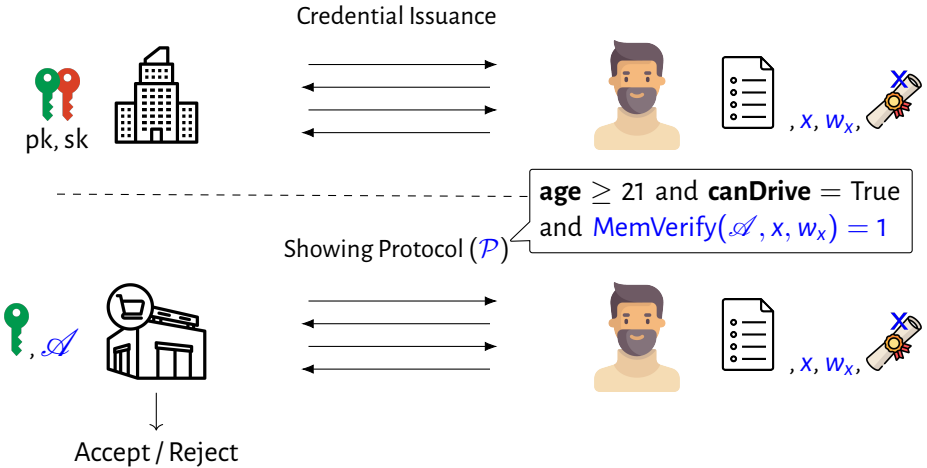
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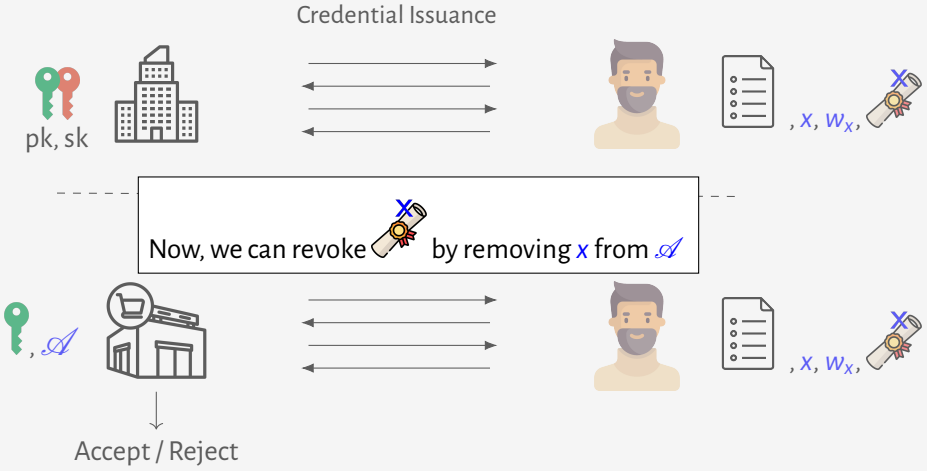
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Prior works on Positive Dynamic Accumulators

Scheme	Assumption	$\lvert w \rvert$	$\lvert \text{upmsg} \rvert_{\text{Add}}$	$\lvert \text{upmsg} \rvert_{\text{Del}}$	$\lvert \text{pp} \rvert$
[CL02 b; LLX07 ; KL24]	Strong RSA	$\ell \cdot \text{poly}(\lambda)$	ℓ^*	ℓ	$\text{poly}(\lambda)$
[BCD+17 ; KL24]	Strong RSA	$\ell \cdot \text{poly}(\lambda)$	–	ℓ	$\text{poly}(\lambda)$
[Ngu05 ; ATS+09 ; CKS09]	q-Strong DH	$\text{poly}(\lambda)$	ℓ^*	ℓ	$s \cdot \text{poly}(\lambda)$
[KB21 ; JML24]	q-Strong DH	$\text{poly}(\lambda)$	–	ℓ	$\text{poly}(\lambda)$
[PST+13 ; YAY+18 ; LLN+23]	M-SIS	$\text{poly}(\lambda) \cdot \log s$	$\text{poly}(\lambda) \cdot \log s^*$	$\text{poly}(\lambda) \cdot \log s$	$\text{poly}(\lambda)$
[ZYH24]	M-SIS	$\text{poly}(\lambda)$	$\text{poly}(\lambda)^*$	$\text{poly}(\lambda)$	$\text{poly}(\lambda) \cdot s \log s$
[CP23]	M-SIS	$\ell \cdot \text{poly}(\lambda)$	ℓ^*	ℓ	$\ell \cdot \text{poly}(\lambda)$
[CP23]+ [WW23]	ℓ-Succinct M-SIS	$\text{poly}(\lambda)$	ℓ^*	ℓ	$\ell^2 \cdot \text{poly}(\lambda)$
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- ℓ: Input's bit length
- $*$: $\lvert \text{upmsg} \rvert = 0$ for a fix set in pre-processing
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Digital Signature

Let $\Sigma = (\text{Gen}, \text{Sign}, \text{Verify})$ be a digital signature

- $\text{Gen}(1^\lambda) \to (\text{pk}, \text{sk})$
- $\text{Sign}(\text{sk}, m) \to \sigma$
- $\text{Verify}(\text{pk}, m, \sigma) \to 1/0$

Security

It should be hard for an adversary to generate (m^*, σ^*) given pk and $\{(m_i, \sigma_i)\}$ where $m^* \neq m_i$ for all i.

Positive Dynamic Accumulator from Digital Signature

Let $\Sigma = (\mathsf{Gen}, \mathsf{Sign}, \mathsf{Verify})$ be a digital signature. In addition, suppose Σ supports the following operations:

- $\mathsf{UpdatePK}(\mathsf{pk}, \mathsf{sk}, \bar{m}) \rightarrow (\mathsf{pk}', \mathsf{upmsg})$
- $\mathsf{UpdateSig}(m, \sigma_m, \mathsf{upmsg}) \rightarrow \sigma'_m$

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Desiderata

- $\mathsf{Verify}(\mathsf{pk}', m, \sigma'_m) = 1$ with overwhelming probability for any $m \neq \bar{m}$
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UpdatePK allows to *revoke* signatures on messages.

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This construction is communication efficient, i.e., $|\mathsf{upmsg}| = O(\#\mathsf{Del})$.

Gadget Matrix

[MP12]

Let $R_q \supseteq \mathbb{Z}_q$ be a ring such that R_q^m admits an ℓ_∞-norm

$$\mathbf{G} = \begin{bmatrix} 1, 2, 4, \ldots, 2^{k-1} & & & \\ & 1, 2, 4, \ldots, 2^{k-1} & & \\ & & \ddots & \\ & & & 1, 2, 4, \ldots, 2^{k-1} \end{bmatrix} \in R_q^{n \times nk}$$

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- $k = \lceil \log q \rceil$.
- There exists a *decomposition* function $\mathbf{G}^{-1} : R_q^n \to R_q^{nk}$ such that for any $\mathbf{u} \in R_q^n$, we have $\mathbf{G} \cdot \mathbf{G}^{-1}(\mathbf{u}) = \mathbf{u}$ and $\|\mathbf{G}^{-1}(\mathbf{u})\|_\infty = 1$

Homomorphic Operations on Matrices

[GSW13 ; BGG+14 ; CP23]

For any $\ell \in \mathbb{N}$, let $\mathcal{F} = \{f_i : \{0,1\}^\ell \to \{0,1\}\}_{i \in \mathbb{N}}$ be a family of Boolean circuits. Then, there exist efficient algorithm EvalF and EvalFX such that for any $\mathbf{B} \in R_q^{n \times \ell m}$, $f \in \mathcal{F}$, and $x \in \{0,1\}^\ell$:

- $\mathsf{EvalF}(f, \mathbf{B}) \to \mathbf{B}_f$
- $\mathsf{EvalFX}(f, \mathbf{B}, x) \to \mathbf{H}_{f,x}$ with $\|\mathbf{H}_{f,x}\|_\infty = 1$

$$\text{s.t.} \quad (\mathbf{B} - x \otimes \mathbf{G}) \cdot \mathbf{H}_{f,x} = \mathbf{B}_f - f(x) \cdot \mathbf{G}$$

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- $\mathsf{EvalF}(f, \mathbf{B}) \to \mathbf{B}_f$
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$$\text{s.t.} \quad (\mathbf{B} - x \otimes \mathbf{G}) \cdot \mathbf{H}_{f,x} = \mathbf{B}_f - f(x) \cdot \mathbf{G}$$

Homomorphic Operations on Matrices

[GSW13 ; BGG+14 ; CP23]

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$$\mathcal{F}_{\mathsf{Indicator}} : \{\mathbb{1}_y : \{0,1\}^\ell \to \{0,1\}\}, \text{ where } \mathbb{1}_y(x) = \begin{cases} 1 \text{ if } x = y \\ 0 \text{ otherwise} \end{cases}$$

Our Construction

Communication efficient accumulator

$$\mathsf{pp} = (\mathbf{A} \in R_q^{n \times \bar{m}}, \mathbf{B} \in R_q^{n \times \ell m}), \mathsf{sk} = \mathbf{T}_{\mathbf{A}}, \mathscr{A}_0 \leftarrow\!\!\$\ R_q^{n \times m}$$

sk *allows to compute a low-norm matrix* $\mathbf{V} \leftarrow \mathsf{SamplePre}_{\mathsf{sk}}([\mathbf{A} \mid \bar{\mathbf{B}}], \mathbf{U})$ *s.t.* $[\mathbf{A} \mid \bar{\mathbf{B}}] \cdot \mathbf{V} = \mathbf{U}$ *for any* $\bar{\mathbf{B}}$.

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- $\mathsf{Add}(\mathsf{pp}, \mathsf{sk}, \mathscr{A}, x)$:
  1. Sample $\mathbf{S}_x \leftarrow \mathsf{SamplePre}_{\mathsf{sk}}([\mathbf{A} \mid \mathbf{B} - x \otimes \mathbf{G}], \mathscr{A})$
  2. Return $\mathbf{S}_x$ as w_x

Agrawal-Boneh-Boyen [ABB10] signature

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- $\mathsf{MemVerify}(\mathsf{pp}, \mathscr{A}, x, w_x)$:
  1. Check if $[\mathbf{A} \mid \mathbf{B} - x \otimes \mathbf{G}] \cdot w_x = \mathscr{A}$ and $\|w_x\|_\infty$ is small

Our Construction

Communication efficient accumulator – Correctness

Let $x \in \{0,1\}^\ell$ with an updated witness w'_x that was generated after deleting $y \neq x \in \{0,1\}^\ell$.
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[\mathbf{A} \mid \mathbf{B} - x \otimes \mathbf{G}] \left(\mathbf{S}_x + \begin{bmatrix} \mathbf{0} \\ \mathbf{H}_{\mathbb{1}_y,\mathbf{B},x} \end{bmatrix} \right) &= \mathscr{A} + (\mathbf{B} - x \otimes \mathbf{G}) \cdot \mathbf{H}_{\mathbb{1}_y,\mathbf{B},x} \\
&= \mathscr{A} + \mathbf{B}_{\mathbb{1}_y} - \mathbb{1}_y(x)\mathbf{G} \\
&= \mathscr{A}' \quad (\text{Since } \mathbb{1}_y(x) = 0)
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- $\|w'_x\|_\infty = \|w_x\|_\infty + \left\|\mathbf{H}_{\mathbb{1}_y,\mathbf{B},x}\right\|_\infty = \|w_x\|_\infty + 1$

By setting the *noise* budget accordingly, we can support poly deletions.

Our Construction

Communication efficient accumulator – Instantiation

Scheme	q	#Add	#Del	$\lvert w_{\mathbf{x}} \rvert$	$\lvert \text{upmsg} \rvert_{\text{Add}}$	$\lvert \text{upmsg} \rvert_{\text{Del}}$	$\lvert \mathscr{A} \rvert$	$\lvert \text{pp} \rvert$
[CP23] (M-SIS)	$\approx 2^{90}$	2^{32}	2^{32}	12MB	4 B	4 B	45KB	14.2MB
[CP23]+[WW23] (ℓ-Succinct M-SIS)	$\approx 2^{150}$	2^{32}	2^{32}	5.5MB	4 B	4 B	75KB	77.3MB
Our work (M-SIS)	$\approx 2^{100}$	—	2^{32}	14.72MB	—	4 B	50KB	16.7MB
Our work (ℓ-Succinct M-SIS)	$\approx 2^{162}$	—	2^{32}	9.33MB	—	4 B	81KB	171.7MB

Our Construction

Communication efficient accumulator – Instantiation

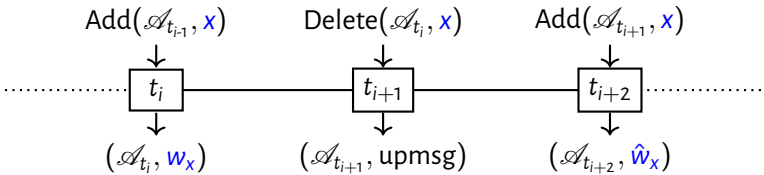
Scheme	q	#Add	#Del	$\lvert w_{\mathbf{x}} \rvert$	$\lvert \text{upmsg} \rvert_{\text{Add}}$	$\lvert \text{upmsg} \rvert_{\text{Del}}$	$\lvert \mathscr{A} \rvert$	$\lvert \text{pp} \rvert$
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- **Replacement-free condition**: Cannot re-add x after it was deleted.

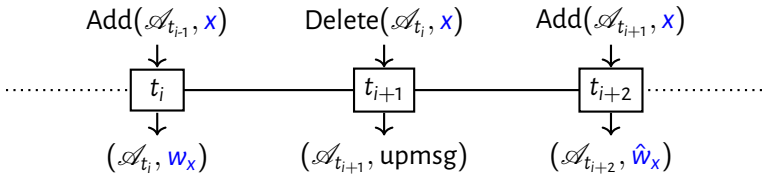
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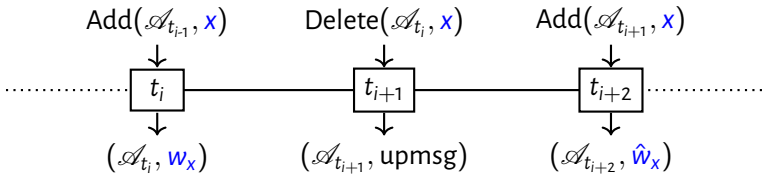
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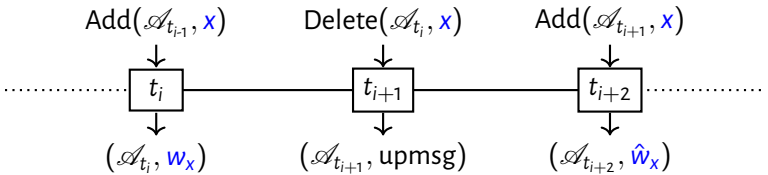
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Theorem

If the replacement-free condition holds and the (module) Short Integer Solution problem is hard, then our construction is a selectively secure communication efficient positive dynamic accumulator.*

Security Analysis

Short Integer Solution (n, m, β)

Given $\bar{\mathbf{A}} \leftarrow\!\!\$\ R_q^{n \times m}$, find $\mathbf{v} \neq \mathbf{0}$ such that $\|\mathbf{v}\| \leq \beta$ and

- $\bar{\mathbf{A}}\mathbf{v} = \mathbf{0}$, for the homogeneous case.
- $\bar{\mathbf{A}}\mathbf{v} = \mathbf{t}$, for the inhomogeneous case w.r.t target $\mathbf{t} \neq 0$.

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Case 1: x^* was never added to the accumulator.
Then $[\mathbf{A} \mid \mathbf{B} - x^* \otimes \mathbf{G}] \cdot w_{x^*} = \mathscr{A}$.
Since w_{x^*} is *short*, it is an inhomogeneous solution for $[\mathbf{A} \mid \mathbf{B} - x^* \otimes \mathbf{G}]$.

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Case 2: x^* was added then remove from the accumulator.
Then there exists $\tilde{w}_{x^*} \neq w_{x^*}$ such that $[\mathbf{A} \mid \mathbf{B} - x^* \otimes \mathbf{G}] \cdot \tilde{w}_{x^*} = \mathscr{A} - \mathbf{G}$.

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Hence, using $(w_{x^*} - \tilde{w}_{x^*})$ we can sample a short $\mathbf{v} \neq \mathbf{0}$ and $[\mathbf{A} \mid \mathbf{B} - x^* \otimes \mathbf{G}]\mathbf{v} = \mathbf{0}$

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Note: Under the replacement-free condition, these two cases are sufficient.

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Non-adaptively secure Positive Dynamic Accumulator + *Adaptively secure Digital signature* $\xRightarrow{[BCD+17]}$ *Adaptively secure Positive Dynamic Accumulator*

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Theorem

Selectively secure *Positive* *Replacement-free* *Accumulator* *(Communication efficient)* + *Adaptively secure* *Digital signature* $\overset{[BCD+17]}{\Longrightarrow}$ *Adaptively secure* *Positive Dynamic* *Accumulator* *(Communication efficient)*

Security Analysis

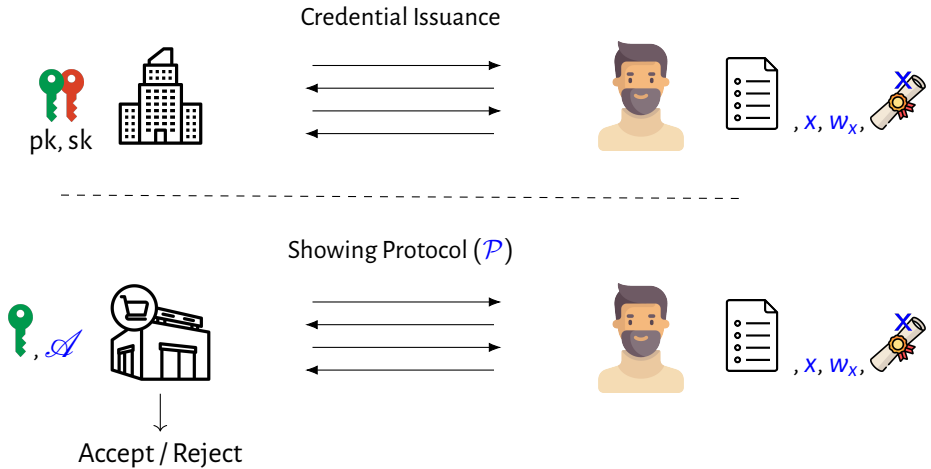
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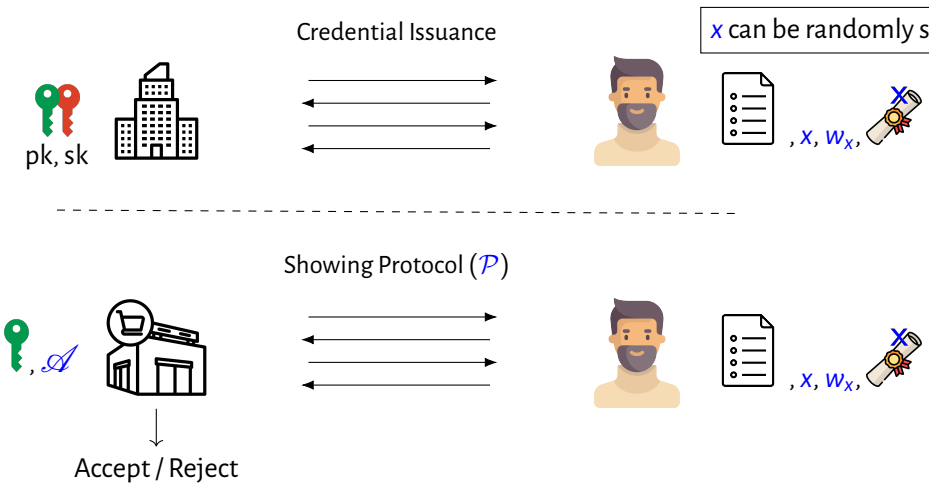
Selectively secure Positive Replacement-free Accumulator (Communication efficient) + *Adaptively secure Digital signature* $\xRightarrow{[BCD+17]}$ *Adaptively secure Positive Dynamic Accumulator (Communication efficient)*

Note: A replacement-free selectively secure accumulator is sufficient for Anonymous Credential Revocation.

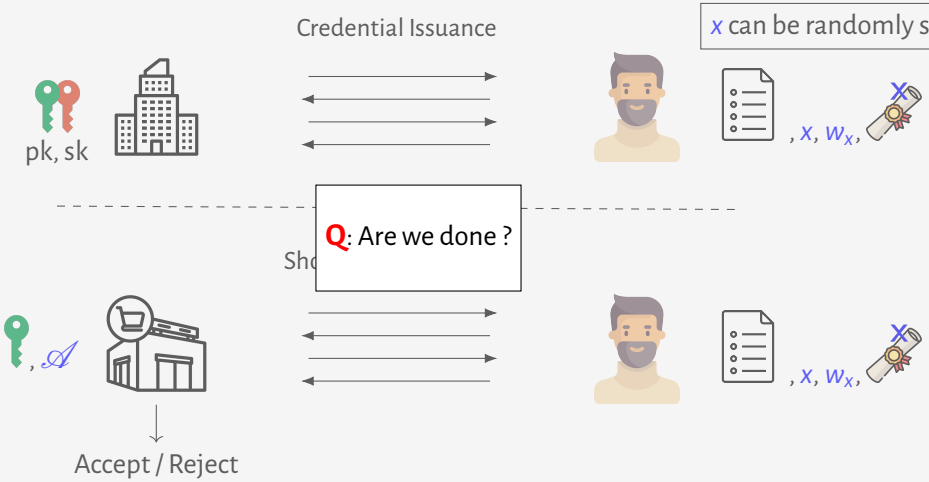
Replacement-free Selectively Secure Accumulator in ACs revocation



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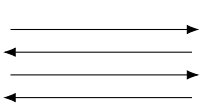
Replacement-free Selectively Secure Accumulator in ACs revocation



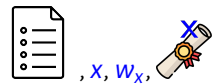
Replacement-free Selectively Secure Accumulator in ACs revocation



Credential Issuance





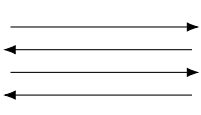


x can be randomly sampled .

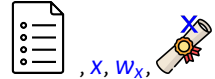
Showing Protocol ($\mathcal{P}$)



Accept / Reject







During the Showing Protocol, we need to prove knowledge of x and w_x s.t. MemVerify$(\mathscr{A}, x, w_x) = 1$.

Replacement-free Selectively Secure Accumulator in ACs revocation

From Lattice-based zero-knowledge proofs [Lyu12 ; ENS20 ; LNP+21 ; LNP22 ; BS23], we know how to prove knowledge of $\mathbf{v}$ such that

$$\mathbf{Cv} = \mathbf{t}, \quad \|\mathbf{v}\| \leq \beta$$

Replacement-free Selectively Secure Accumulator in ACs revocation

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For our construction, we need to prove knowledge of (x, w_x) such that

$$[\mathbf{A} \mid \mathbf{B} - x \otimes \mathbf{G}] \cdot w_x = \mathscr{A}, \quad \|w_x\| \leq \beta' \tag{1}$$

Replacement-free Selectively Secure Accumulator in ACs revocation

From Lattice-based zero-knowledge proofs [Lyu12 ; ENS20 ; LNP+21 ; LNP22 ; BS23], we know how to prove knowledge of $\mathbf{v}$ such that

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How can we handle x?

- Compute a commitment $\mathsf{Com}(x; r)$ and produce a proof $\pi_{\mathsf{Com}} = (\mathbf{w}, c, z)$.
- From z, we can extract $z_x = y_x + c \cdot x$ such that

$$[c\mathbf{A} \mid c\mathbf{B} - z_x \otimes \mathbf{G}] \cdot w_x = c \underbrace{[\mathbf{A} \mid \mathbf{B} - x \otimes \mathbf{G}] \cdot w_x}_{\mathscr{A}} + [\mathbf{0} \mid -y_x \otimes \mathbf{G}] \cdot w_x$$

Thank You!

https://ia.cr/2025/1099



Some icons were sourced from Flaticon.com

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